Quantum Theory Contributions

- • Introduced the idea of quantized energy
- • Used quantum rules to explain atomic stability
- • Influenced development of quantum mechanics
- • Guided scientists like Heisenberg & Schrödinger
- • Key figure in Copenhagen Interpretation

Complementarity Principle

- • Proposed dual nature of matter: wave + particle
- • Both properties cannot be observed at same time
- • Observation depends on measurement setup
- • Fundamental philosophical idea of quantum mechanics

Career & Achievements

- • Worked with Ernest Rutherford on atomic research
- • Debated quantum ideas with Albert Einstein
- • Nobel Prize in Physics (1922)
- • Contributed to nuclear fission understanding
- • Encouraged peaceful use of nuclear energy

Legacy

- • Founded Institute of Theoretical Physics in Copenhagen
- • Mentored future Nobel laureates
- • Major contributor to modern quantum physics
- • His ideas influenced science, electronics & chemistry
- • Remembered as one of the greatest physicists

Early Life

- • Born in 1885 in Copenhagen, Denmark
- • Grew up in an academic family; father was a physiology professor
- • Studied physics at the University of Copenhagen
- • Early interest in atomic structure and scientific research
- • Worked in England with leading physicists

Bohr Model of the Atom (1913)

- • Electrons revolve in fixed circular orbits called energy levels
- • Electrons do not radiate energy while in stable orbits
- • Energy absorbed when electron jumps to higher orbit
- • Energy emitted as photon when electron drops to lower orbit
- • Successfully explained hydrogen spectrum